

Project 1 — E-Commerce Sales Intelligence Dashboard

WEEK 2

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Week 2 Objective

Transform the Week 1 data pipeline output into a fully interactive business intelligence product. Build a 3-page Power BI dashboard with DAX measures and a live-deployed Streamlit app accessible via a public URL. By end of Week 2, the project has a resume-ready GitHub repo, a Power BI .pbix file, and a permanently live interactive dashboard at sudiksha-ecommerce-analytics.streamlit.app.

Final deliverables — Week 2

Power BI Dashboard: 3 pages — Revenue Overview, Delivery & Operations, Customer Intelligence

DAX Measures: 10 measures including YoY, MTD, Rolling 30D, Late Rate %, RFM KPIs

Streamlit App (LIVE): <https://sudiksha-ecommerce-analytics.streamlit.app>

GitHub Repo (LIVE): <https://github.com/sudiksha14/ecommerce-sales-dashboard>

README updated with live demo link, business findings, and how-to-run instructions

Day-by-day summary

Week 2 Day 1 Power BI — Data model + star schema

- | | | |
|---|---|------|
| 1 | Loaded all 6 processed CSVs into Power BI Desktop via Get Data → Text/CSV | Done |
| 2 | Set correct data types: Decimal for revenue columns, Whole Number for keys, Date for full_date | Done |
| 3 | Built star schema in Model view: 5 relationships (fact_orders → dim_date, dim_customer, dim_product, dim_seller, dim_geography) | Done |
| 4 | Verified all relationships show correct 1-to-many cardinality with no errors | Done |

Week 2 Day 2 Power BI — 10 DAX measures

- | | | |
|---|---|------|
| 1 | Created Total Revenue, Total Orders, Avg Order Value, Avg Delivery Days, Avg Review Score | Done |
| 2 | Created Late Order Rate % using DIVIDE + CALCULATE with is_late = 1 filter | Done |
| 3 | Created Revenue YoY % using SAMEPERIODLASTYEAR time intelligence function | Done |
| 4 | Created Rolling 30D Revenue using DATESINPERIOD with -30 DAY window | Done |
| 5 | Created Late Rate Status using SWITCH(TRUE()) returning Good/Watch/Critical | Done |
| 6 | Created Revenue MTD using DATESMTD time intelligence function | Done |
| 7 | Formatted all measures with correct display formats (currency, %, decimal) | Done |

Week 2 Day 3 Power BI — Page 1: Revenue Overview

1	Added 5 KPI cards: Total Revenue (R\$16.01M), Total Orders (99K), Avg Order Value (R\$160.99), Revenue MTD, Revenue YoY %	Done
2	Built monthly revenue line chart with dim_date[year] + dim_date[month] on X-axis	Done
3	Built top 10 categories horizontal bar chart with Top N filter	Done
4	Built revenue by day of week column chart — fixed sort order Mon→Sun via DayNum column	Done
5	Built price bucket donut chart with 4 segments	Done
6	Added page title text box and year slicer	Done
Week 2 Day 4 Power BI — Page 2: Delivery & Operations		
1	Added 4 KPI cards: Avg Delivery Days (12.5), Late Order Rate % (7.9%), Late Rate Status (Watch), Avg Review Score (4.09)	Done
2	Built delivery days by state horizontal bar — sorted descending, constant line at national avg	Done
3	Built monthly late order rate line chart with avg reference line	Done
4	Built delivery days distribution column chart (bell curve) filtered to 0-60 days	Done
5	Built on-time vs late pie chart — added Delivery Status calculated column (0→On-Time, 1→Late)	Done
6	Added year slicer for page-level filtering	Done
Week 2 Day 5 Power BI — Page 3: Customer Intelligence		
1	Loaded rfm_segments table from sql/result_03_rfm.csv	Done
2	Created Champions Count and At Risk Count DAX measures in rfm_segments table	Done
3	Created Total Customers = COUNTROWS(rfm_segments)	Done
4	Added RFM segment tile slicer — all 8 segments visible	Done
5	Built revenue by RFM segment bar chart with color-coded segments	Done
6	Built review score distribution bar chart using review_score_rounded column	Done
7	Built payment type donut chart and RFM segment count bar chart	Done
8	Added state map visual — set state column data category to State or Province	Done
9	Added Top 20 customers detail table	Done
Week 2 Day 6 Streamlit App — Build + local test		
1	Built full 3-page Streamlit app (streamlit_app.py) mirroring Power BI dashboard	Done
2	Page 1: 5 KPIs, monthly trend, top 10 categories, price bucket donut, day of week, AOV	Done
3	Page 2: 4 KPIs, delivery by state with reference line, monthly late rate, histogram, pie	Done
4	Page 3: 4 KPIs, RFM multiselect filter, segment revenue, review scores, payment types, top 20 table	Done
5	Added sidebar with Year/Category/State filters affecting all pages	Done
6	Created .streamlit/config.toml for light theme configuration	Done
7	Tested locally at localhost:8501 — all 3 pages and filters working correctly	Done

Week 2 Day 7 Streamlit App — Deploy to Streamlit Cloud

1	Created requirements.txt with streamlit, plotly, pandas, numpy	Done
2	Pushed streamlit_app.py, requirements.txt, .streamlit/config.toml to GitHub	Done
3	Pushed data/processed/ and sql/ result CSVs to GitHub for cloud access	Done
4	Deployed on share.streamlit.io — selected repo, branch main, main file streamlit_app.py	Done
5	Set custom URL: sudiksha-ecommerce-analytics.streamlit.app	Done
6	Updated README.md with live demo link	Done
7	Pushed PDFs/ and Power BI Report/ folders to GitHub	Done
8	Added live URL to resume bullet points and LinkedIn Featured section	Done

All challenges faced in Week 2 + fixes

Day	Challenge	Root cause	Fix applied
DAY 1	dim_date[full_date] loaded as text not Date type	Power BI inferred full_date as text from CSV. Time intelligence DAX functions require a proper Date column.	Went to Transform Data → dim_date → full_date column → changed data type to Date. Clicked Close & Apply.
DAY 2	Late Order Rate % showing 787.1% instead of 7.9%	DAX measure had * 100 at the end, but Power BI format was also set to Percentage — double multiplication gave 787%.	Removed * 100 from the measure. Set format to Percentage in Measure tools ribbon. Power BI handles x100 display automatically.
DAY 2	Revenue YoY % returned blank for most periods	SAMEPERIODLASTYEAR requires a full calendar year of prior data. Dataset starts mid-2016 so many periods have no prior year.	Expected behaviour — documented. Added a note in the report tooltip explaining the measure requires prior year data.
DAY 3	Day of week X-axis in wrong order (Sunday before Saturday)	Power BI sorted day_of_week alphabetically by default. No built-in weekday sort.	Added DayNum calculated column in Power Query (Mon=1...Sun=7). Set day_of_week Sort By Column to DayNum in Model view.
DAY 3	All KPI cards showing same green color	Default Power BI card visual background applied same color to all cards.	Clicked each card individually → Format visual → Colors → assigned unique color per card (purple, teal, coral, amber, green).
DAY 4	Late Rate Status card showing 'Watch' in small text	Card visual callout value font was too small for text measures.	Clicked card → Format → Callout value → increased font size to 24 and set bold.
DAY 5	State map showing world map instead of Brazil	dim_customer[state] column had no geographic data category set — Power BI did not know it represented Brazilian states.	Data view → dim_customer → state column → Column tools → Data category → State or Province. Map auto-zoomed to Brazil.

DAY 5	Review score chart showing one bar instead of 1-5 distribution	avg_review_score is a decimal (e.g. 4.2) — using it directly on X-axis created too many unique values.	Added review_score_rounded column in Power Query using Number.Round([avg_review_score], 0). Used this column on X-axis.
DAY 5	RFM slicer only showing 2 tiles (At Risk, Champions)	Tile style slicer was too small to display all 8 segments — overflow was hidden.	Changed slicer style from Tile to List. All 8 segments visible. Resized slicer panel to fit all items.
DAY 6	Streamlit KPI cards invisible in dark mode	White card background (#FFFFFF) was invisible against Streamlit's dark gray default background.	Created .streamlit/config.toml with base='light' theme. App reloaded in light mode with all cards clearly visible.
DAY 7	config.toml caused UnicodeDecodeError on Streamlit Cloud	PowerShell echo command saved config.toml as UTF-16 encoding. Streamlit Cloud's Python 3.14 could not decode it.	Deleted the file. Recreated it manually in VS Code (New File → paste content → Ctrl+S). Confirmed UTF-8 encoding shown at VS Code status bar. Pushed and redeployed.
DAY 7	git add .streamlit/config.toml failed — pathspec did not match	.streamlit/config.toml file did not exist yet when first git add was attempted.	Created the .streamlit/ folder and config.toml file first, then ran git add successfully.
DAY 7	ecommerce-sales-dashboard subdomain already taken on Streamlit Cloud	Another user had already registered that subdomain on share.streamlit.io.	Used sudiksha-ecommerce-analytics as the subdomain. Final URL: sudiksha-ecommerce-analytics.streamlit.app

All 10 DAX measures — quick reference

Measure	Pattern	Format	Result
Total Revenue	SUM(fact_orders[total_payment])	Currency	R\$16.01M
Total Orders	COUNTROWS(fact_orders)	Whole number	99,441
Avg Order Value	DIVIDE(SUM(total_payment), COUNTROWS(fact_orders), 0)	Currency	R\$160.99
Late Order Rate %	DIVIDE(CALCULATE(COUNTROWS, is_late=1), COUNTROWS, 0)	Percentage	7.9%
Avg Delivery Days	AVERAGE(fact_orders[delivery_days])	Decimal 1dp	12.5
Avg Review Score	AVERAGE(fact_orders[avg_review_score])	Decimal 2dp	4.09
Revenue YoY %	DIVIDE(Current - SAMEPERIODLASTYEAR, SAMEPERIODLASTYEAR, 0) * 100	Percentage	Varies
Rolling 30D Revenue	CALCULATE(SUM, DATESINPERIOD(full_date, LASTDATE, -30, DAY))	Currency	Varies
Late Rate Status	SWITCH(TRUE(), <=5 Good, <=10 Watch, Critical)	Text	Watch
Revenue MTD	CALCULATE(SUM(total_payment), DATESMTD(dim_date[full_date]))	Currency	Varies

Key learnings from Week 2

1. Power BI time intelligence functions (SAMEPERIODLASTYEAR, DATESMTD, DATESINPERIOD) require a properly typed Date column — always verify full_date is Date not Text after loading.
2. DAX percentage measures should NOT multiply by 100 when the measure format is set to Percentage — Power BI does the x100 display automatically.
3. Sort By Column is the correct Power BI pattern for custom axis ordering (weekdays, months) — create a numeric sort key column in Power Query.
4. Power BI data categories (State or Province, City) must be set manually on geographic columns — without them, map visuals fall back to world-level geocoding.
5. Streamlit config.toml must be saved as UTF-8 — PowerShell echo creates UTF-16 files which crash Python 3.14's codec on Streamlit Cloud.
6. SWITCH(TRUE(), condition1, result1, ...) is the cleanest DAX pattern for multi-condition KPI flag measures — more readable than nested IF statements.
7. Streamlit @st.cache_data decorator is essential for dashboard apps — without it, all 6 CSVs reload on every user interaction causing slow performance.
8. A live deployed URL (streamlit.app) on a resume is worth more than a static screenshot — it proves the ability to build AND ship, not just analyse.
9. RFM segmentation results (from SQL) can be loaded as a separate table in both Power BI and Streamlit without needing a formal relationship — useful for cross-tool consistency.
10. Plotly charts in Streamlit support add_vline() and add_hline() for reference lines — replicating the Power BI analytics constant line feature exactly.

Complete Project 1 summary — Weeks 1 + 2

Day	Focus	Deliverable	Status
Week 1 Day 1	Environment setup	Anaconda, VS Code, 9 Olist CSVs verified	Done
Week 1 Day 2	ETL pipeline	6 clean star-schema tables, 253k rows total	Done
Week 1 Day 3	EDA — 15 charts	4 sections, real business insights in every title	Done
Week 1 Day 4	Advanced SQL	Rolling revenue, cohort retention, RFM (DuckDB)	Done
Week 1 Day 5	GitHub repo	Public repo + professional README live	Done
Week 2 Day 1	Power BI model	Star schema, 5 relationships, correct data types	Done
Week 2 Day 2	DAX measures	10 measures: time intelligence, KPI flags, rolling	Done
Week 2 Day 3	Page 1 — Revenue	5 KPIs, line chart, top 10 categories, donut, DOW	Done
Week 2 Day 4	Page 2 — Delivery	4 KPIs, state bars, late rate trend, histogram, pie	Done
Week 2 Day 5	Page 3 — Customer	RFM slicer, segment revenue, review, payment, map	Done
Week 2 Day 6	Streamlit app built	3-page app, sidebar filters, Plotly charts, local test	Done

Week 2 Day 7	Streamlit app deployed	Live at sudiksha-ecommerce-analytics.streamlit.app	Done
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Live App URL: <https://sudiksha-ecommerce-analytics.streamlit.app>

GitHub URL: <https://github.com/sudiksha14/ecommerce-sales-dashboard>

Resume line: Built end-to-end BI pipeline on 99,441 orders (R\$16M GMV) — ETL, star schema, DAX, Streamlit (live URL)

Next: Project 2 — Customer Churn Prediction App (Python + scikit-learn + Streamlit)